Financial KPI Analysis for a Startup – SaaS Customer Analytics Dashboard

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Abstract

This project focuses on analyzing key business and customer metrics for a Software-as-a-Service (SaaS) company using SQL and Tableau. The objective was to transform raw customer and revenue data into meaningful insights that support business decision-making. SQL was used to calculate key performance indicators (KPIs), while Tableau was used to create an interactive dashboard that visualizes customer activity, revenue performance, churn rate, and customer acquisition costs. The resulting dashboard provides a clear overview of the company's performance and highlights important trends affecting growth and profitability.

Introduction

Data analytics plays a critical role in helping SaaS companies understand customer behavior, monitor revenue growth, and evaluate business performance. This project was developed to analyze customer and revenue data and present the results through a professional business intelligence dashboard.

The dashboard focuses on several important SaaS metrics, including active customers, churn rate, monthly revenue trends, customer distribution by subscription plan, revenue generated by each plan, and customer acquisition cost (CAC). These metrics provide valuable insights into customer retention, revenue generation, and operational efficiency.

Tools Used

- **MySQL** for data storage and SQL analysis
- **Tableau Public** for data visualization and dashboard development
- **Microsoft Excel/CSV Files** for dataset management

Methodology

The project followed a structured data analysis process:

1. Customer and revenue datasets were imported into MySQL.
2. SQL queries were written to calculate key business metrics.
3. Queries were used to generate KPI outputs such as total revenue, active customers, churn rate, customer counts by plan, revenue by plan, and average customer acquisition cost.
4. The datasets were connected to Tableau.
5. Multiple visualizations were created and combined into a single interactive dashboard.
6. The dashboard was formatted to improve readability and presentation quality.

Key Findings

The analysis revealed several important business insights:

- The company generated a total revenue of approximately **249,800**.
- There were **832 active customers** across all subscription plans.
- The overall **churn rate was 16.8%**, indicating that a portion of customers had discontinued their subscriptions.
- The **Enterprise plan generated the highest revenue**, contributing approximately **167,000**, making it the most valuable subscription tier.
- The **Pro plan recorded the highest number of customers**, while Enterprise customers produced greater revenue per customer.
- The **Enterprise plan also had the highest average customer acquisition cost (CAC)**, suggesting that acquiring high-value customers requires greater investment.
- Monthly revenue showed a consistent upward trend throughout 2024 and reached its highest levels during early 2025 before experiencing a decline in later months.

Conclusion

This project demonstrates how SQL and Tableau can be combined to perform business analysis and create meaningful visualizations. The dashboard provides a comprehensive overview of customer performance, revenue growth, customer retention, and acquisition efficiency within a SaaS business environment.

The results highlight the importance of monitoring key performance indicators to support strategic decision-making. By identifying revenue trends, customer distribution, churn levels, and acquisition costs, businesses can make data-driven decisions to improve growth, profitability, and customer retention.